

Ranjan Bera

AWS DEVOPS ENGINEER

PROFILE SUMMARY

DevOps Engineer with over 4 years of experience designing, automating, and managing cloud-native infrastructure on AWS. Proven expertise in Kubernetes (EKS), Terraform, CI/CD, GitOps, infrastructure as code, cloud security, observability, and automation. Hands-on experience implementing Generative AI and LLMOps solutions using Amazon Bedrock and self-hosted LLM platforms (Ollama). Skilled in building scalable, secure, and highly available platforms while optimizing cost, reliability, and developer productivity, and collaborating across development, security, and operations teams to implement DevSecOps best practices.

WORK EXPERIENCE

AWS DevOps Engineer

TATA Consultancy Services Ltd.

Dec 2021 - Present

- Designed, implemented, and maintained cloud-native infrastructure on AWS supporting enterprise-scale applications.
- Managed Kubernetes (EKS) clusters ensuring high availability, scalability, security, and operational excellence.
- Developed reusable Terraform modules for VPC, IAM, S3, EKS, networking, and platform services following AWS best practices.
- Built and optimized CI/CD pipelines using Jenkins, GitLab CI/CD, Helm, and ArgoCD to improve deployment efficiency.
- Implemented GitOps workflows for Kubernetes deployments to reduce manual intervention and improve reliability.
- Automated infrastructure provisioning and application deployment using Infrastructure as Code principles.
- Designed and implemented secure access patterns using IAM Roles for Service Accounts (IRSA) and AWS Secrets Manager.
- Built serverless automation solutions using AWS Lambda, Python, SNS, SQS, and EventBridge.
- Implemented centralized authentication using OAuth2 Proxy integrated with GitLab.
- Optimized infrastructure costs using Spot Instances, Auto Scaling Groups, and Karpenter.
- Developed monitoring, alerting, and observability solutions using CloudWatch, Grafana, and Prometheus.
- Implemented automated URL monitoring and notification systems for production environments.
- Designed CloudFront-based maintenance and failover solutions for high availability.
- Developed automated AWS Secrets Manager rotation frameworks using Lambda and Terraform.
- Collaborated with development, security, and operations teams to implement DevSecOps best practices.
- Contributed to GenAI & LLMOps initiatives by designing and deploying Generative AI solutions using Amazon Bedrock and building local LLM environments with Ollama, enabling RAG proof-of-concepts and streaming response architectures.

PROJECTS

Enterprise GenAI Platform using Amazon Bedrock

- Developed a secure Generative AI platform leveraging Amazon Bedrock foundation models.
- Integrated Bedrock APIs with enterprise applications for conversational AI and



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4 Years of experience

KEY SKILLS

- ✓ Kubernetes
- ✓ AWS
- ✓ Terraform
- ✓ Jenkins
- ✓ GitLab CI/CD
- ✓ ArgoCD
- ✓ Docker
- ✓ Helm
- ✓ Python
- ✓ Bash
- ✓ Git
- ✓ Prometheus
- ✓ Grafana
- ✓ AWS Lambda
- ✓ Amazon Bedrock

ACHIEVEMENTS & HIGHLIGHTS

Key Contributions

Contributed to design and delivery of enterprise-grade cloud platforms, reusable Terraform modules, GitOps workflows, and GenAI/LLMOps proof-of-concepts using Amazon Bedrock and Ollama.

EXTRA-CURRICULAR ACTIVITIES

Community Contribution

Participated in knowledge-sharing initiatives and internal tech forums focused on cloud best practices, DevOps automation, and AI/ML infrastructure topics.

knowledge retrieval use cases.

- Implemented IAM-based security controls, monitoring, and cost optimization following AWS Well-Architected principles.
- Technologies: AWS Bedrock, Lambda, API Gateway, IAM, Python, CloudWatch.

Local LLM Platform using Ollama

- Built a self-hosted AI platform using Ollama and open-source Large Language Models.
- Containerized and deployed LLM services using Docker and Kubernetes to enable secure offline AI experimentation.
- Evaluated multiple foundation models for performance, latency, and infrastructure efficiency.
- Technologies: Ollama, Docker, Kubernetes, Python, Linux.

Retrieval-Augmented Generation (RAG) Assistant

- Developed an AI assistant leveraging document ingestion, embeddings, vector search, and LLM inference to improve contextual responses.
- Implemented contextual retrieval mechanisms and streaming response support for real-time interactions.
- Automated deployments using CI/CD pipelines and Infrastructure as Code.
- Technologies: Python, AWS, Kubernetes, Terraform, LLMs.

GitLab Deployment on AWS EKS

- Deployed and managed self-hosted GitLab using Helm on EKS.
- Integrated AWS RDS and Redis for scalable backend services and implemented backup, monitoring, and disaster recovery processes.
- Technologies: GitLab, Helm, EKS, AWS RDS, Redis.

Centralized OAuth2 Proxy Platform

- Implemented centralized authentication for multiple applications using OAuth2 Proxy and GitLab as Identity Provider.
- Secured internal applications including Jenkins and Kibana through unified authentication workflows.
- Technologies: OAuth2 Proxy, GitLab, Kubernetes, NGINX Ingress.

AWS Secrets Rotation Framework

- Developed an automated secrets rotation solution using AWS Lambda and Terraform.
- Integrated secret updates with Kubernetes workloads using IRSA and AWS Secrets Manager to ensure compliance with security best practices.
- Technologies: AWS Lambda, Terraform, Secrets Manager, EKS, Python.

EDUCATION

MS/M.Sc(Science) | Computers

2021

Vidyasagar University

Pass

B.Sc | Computer Science and Technology

2019

Midnapore College, Midnapore

Marks - 64%

CERTIFICATIONS

DevOps Engineer Masters Program

Hands-on approach to AI for real-world applications

LANGUAGES

- English
- Hindi
- Bengali

SOCIAL LINKS

<https://www.linkedin.com/in/ranjanbera1112/>

HOBBIES

- Technical Blogging
- Open Source Contribution
- Trail Running